# Sebastian Villalobos

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ML researcher in mechanistic interpretability and LLM evaluation. 4+ years of applied data science and RLHF evaluation at Scale AI and Welo Data. Wrote an interpretability toolkit from scratch and applied the toolkit to a 3B-parameter LLM to compare behavior installed in activation space vs. in weights. Co-author of a NeurIPS 2026 workshop paper on multilingual LLM safety.

## EDUCATION

**Georgetown University — M.S. in Data Science & Analytics** *Expected May 2027*

- GPA: 4.0/4.0. Merit scholarship. Teaching Assistant: Programming Intro; AI Modeling in Practice; Machine Learning Foundations; Advanced Machine Learning & AI; Stats for Business Analytics; Database Systems & SQL; Applied AI Lab
- Co-designed Applied AI Lab with Prof. Babak Zafari, a non-technical AI course for MBA students

**Universidad Iberoamericana — B.S. Mechatronics & Production Engineering** *Mexico City*

- Tuition scholarship for academic merit

## RESEARCH PROJECTS

**ModelLens — Neural Network Interpretability Toolkit (GitHub)** *Georgetown University — In Progress*

- Built a from-scratch PyTorch interpretability toolkit that auto-detects transformers, CNNs, LSTMs, and MLPs and runs the analyses each supports (patching, circuit discovery, logit lens, sparse autoencoders, probing)
- Shipped as an open-source MCP server with 35 passing tests, exposing each analysis as a callable tool

**Stoic Steering — Intervention Locus & Failure Analysis (GitHub)** *Personal Project — In Progress*

- Generated contrastive-pair training data from a source corpus, plus a stance-balanced forced-choice eval set calibrated to near-chance before any measurement
- Trained LoRA adapters and swept layers and coefficients against activation steering (CAA) under seed-averaged, decoding-matched controls
- Self-identified a decoding confound behind the original positive result (steered outputs sampled, baselines greedy) and published the correction
- Showed that under matched decoding, activation steering had no effect on style, content, or decisions, while LoRA changed decisions; the circuit analysis shows the same ordering

## WORK EXPERIENCE

**Georgetown University, McDonough School of Business — Research Assistant** *Aug 2026 – Present*

- Conducting research on LLM uncertainty quantification with Prof. Babak Zafari

**Welo Data — Data Science & Analytics Engineer** *Nov 2024 – Aug 2025*

- Designed an XGBoost task-difficulty model over 120k+ records, raising annotation accuracy 10% via task allocation
- Created a Gaussian Mixture Model recommender for rater–project matching, raising annotation quality by 10–15%
- Automated a high-volume financial ETL pipeline in Redshift (0.001% error rate), cutting reconciliation time by 70%

**Scale AI — Data Operations Engineer** *Oct 2022 – Nov 2024*

- Developed automated evaluation frameworks for NVIDIA Nemotron's Reinforcement Learning from Human Feedback (RLHF) alignment, using GPT-4 as Judge-LLM and HelpSteer2 methodology to score outputs on helpfulness and safety attributes for reward-model training data
- Built LLM quality linting system adopted across GenAI organization, automating error detection via programmatic flagging and reducing manual review by 40%

**Scale AI — Operations Data Analyst & Specialist** *Feb 2021 – Oct 2022*

- Modeled batch acceptance with logistic regression on 12+ months of operational data, improving approval rates by 20%

## PUBLICATIONS

Alva Jalil, Harper David, Thornton Abigail E., **Villalobos Alva Rafael S.**, et al. (2026). *Global Security Blind Spots: Translation as a Safety Bypass in Low-Resource Languages*. NeurIPS 2026 Workshop on Foundations of Language Model Security (FLMSec).

Villalobos-Alva Jalil, ..., **Villalobos-Alva Sebastian**, et al. (2022). *Protein Science Meets Artificial Intelligence: A Systematic Review*. Front. Bioeng. Biotechnol.

## SKILLS & CERTIFICATIONS

**Mechanistic Interpretability:** Activation & Attribution Patching, Automated Circuit Discovery, Logit Lens, Residual-Stream Analysis, Dictionary-Learning Sparse Autoencoders, Linear Concept Probing
**Model Training & Evaluation:** LoRA Fine-tuning, Supervised Fine-tuning, Contrastive Activation Addition (CAA), Synthetic Dataset Curation, RLHF Evaluation (HelpSteer2), Judge-LLM and Judge-Free Forced-Choice Evaluation
**Programming:** Python (PyTorch, HuggingFace, Pandas, NumPy, Scikit-learn), SQL, NoSQL, TypeScript, R, Git, Linux, Docker
**Languages & Certifications:** Spanish (Native), English (Fluent); MIT MicroMasters in Statistics and Data Science